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Justification:

A number of problems related to classes in Darwin Core were identified and discussed at the Documenting Darwin Core workshop at TDWG 2013. There was an apparent consensus at the meeting regarding steps that should be taken to fix these problems. In particular:

- Darwin Core classes exist in two namespaces, dwc: (<http://rs.tdwg.org/dwc/terms/>) and dwctype: (<http://rs.tdwg.org/dwc/dwctype/>). In some cases, there are parallel terms in both namespaces. To reduce confusion when typing resources, all Darwin Core-defined classes should be in a single namespace.
- The class definitions contain phrases such as "The category of information pertaining to " and "A resource describing " which make it unclear what entities should be considered instances of those classes. The class definitions should define the types of the entities themselves and not categories or records related to the entities.
- Some class definitions are unclear, e.g. are circular or contain ambiguous terms. Class definitions should be clear enough that users will know what entities should be included in the class.
- Concerns about the inclusion of an "Individual" or "Organism" class (Issue 69) seemed to have been allayed and an adequate constituency felt that its inclusion in Darwin Core would be beneficial and would help in clarifying the definitions of some of the ambiguous classes.

An ad hoc group (the submitters of this issue) volunteered or were drafted to turn this consensus into concrete proposals for clarifying the Darwin Core class terms. That group proposes the following general changes to address the problems identified above:

1. All Darwin Core-defined class terms will be defined in the <http://rs.tdwg.org/dwc/terms/> (dwc:) namespace.
2. dcterms:Location (<http://purl.org/dc/terms/Location>), which is imported into Darwin Core from Dublin Core will continue to be used as a class in Darwin Core.
3. The terms in the <http://rs.tdwg.org/dwc/dwctype/> (dwctype:) namespace, which are all class terms, will be deprecated. dwc: namespace analogues will be created for any dwctype: classes that don't already have analogues. This deprecation includes no longer importing dcmitype:Event (<http://purl.org/dc/dcmitype/Event>), whose definition is confusing in the presence of an Occurrence class.
4. Issue 69 will be resolved by the addition of the dwc:Organism class (<http://rs.tdwg.org/dwc/terms/Organism>).
5. Each class term definition will be modified so that it defines the entities included in the class rather than a "category" or data record about the entity. This generally means removing the phrases "The category of information pertaining to " and "A resource describing

" from the definition.

6. Ambiguous class definitions will be modified so that users are clear about what entities are included in them. This may be accomplished using in the definition English or URI terms whose meanings are well-known.

Note: dwctype:MaterialSample is the only Darwin Core class whose normative RDF definition includes a machine-readable (i.e. non-literal) property that links it to another class. It is defined in the RDF as `rdfs:subClassOf OBI:0100051` (http://purl.obolibrary.org/obo/OBI_0100051). Under the current proposal this linkage would be removed since as a dwctype: namespace term, dwctype:MaterialSample would be deprecated and since the dwc: namespace analogue, dwc:MaterialSample does not include this linkage.

To facilitate public discussion about the definitions of particular classes, each class which is new or which has a changed definition will be entered as a separate issue with reference to this issue as the overall rationale for the change. A table which summarizes all of the proposed changes can be viewed at:

http://code.google.com/p/tdwg-rdf/wiki/DwcClassChanges#Summary_of_proposed_changes

class terms (existing and proposed additions)	proposed revised definitions	existing dwc: namespace analogue	existing dwctype: namespace analogue
dwc:Organism	A particular organism or defined group of organisms considered to be taxonomically homogeneous. An organism in the sense used here is defined as OBI:0100026 (http://purl.obolibrary.org/obo/OBI_0100026). Instances of the Organism class are intended to facilitate linking of one or more Identification	N/A	N/A

	instances to one or more Occurrence instances. Therefore, things that are typically assigned scientific names (such as viruses, hybrids, and lichens) and aggregates whose occurrences are typically recorded (such as packs, clones, and colonies) are included in the scope of this class.		
dwc:Occurrence	The existence of a dwc:Organism (http://rs.tdwg.org/dwc/terms/Organism) at a particular place at a particular time.	dwc:Occurrence	dwctype:Occurrence
dwc:MaterialSample	The physical results of a sampling (or subsampling) event. In biological collections, the material sample is typically collected, and either preserved or destructively processed.	dwc:MaterialSample	dwctype:MaterialSample
dwc:Event	An action that occurs at a place and time.	dwc:Event	dcmitype:Event
dcterms:Location	A spatial region or named place.	dcterms:Location	dwctype:Location
dwc:GeologicalContext	Geological information, such as stratigraphy, that qualifies a region or place.	dwc:GeologicalContext	none
dwc:Identification	A taxonomic determination (e.g., the assignment of a scientific name).	dwc:Identification	none
dwc:Taxon	A group of organisms considered by taxonomists to form a homogeneous	dwc:Taxon	dwctype:Taxon

	unit. An organism in the sense used here is defined as OBI:0100026 (http://purl.obolibrary.org/obo/OBI_0100026).		
dwc:ResourceRelationship	A relationship of one <code>rdfs:Resource</code> (http://www.w3.org/2000/01/rdf-schema#Resource) such as an Occurrence, Taxon, Location, etc., to another.	dwc:ResourceRelationship	none
dwc:MeasurementOrFact	A measurement or fact about an <code>rdfs:Resource</code> (http://www.w3.org/2000/01/rdf-schema#Resource) such as an Occurrence, MaterialSample, GeologicalContext, etc.	dwc:MeasurementOrFact	none
dwc:PreservedSpecimen	A preserved specimen.	none	dwctype:PreservedSpecimen
dwc:FossilSpecimen	A fossilized specimen.	none	dwctype:FossilSpecimen
dwc:LivingSpecimen	A living specimen.	none	dwctype:LivingSpecimen
dwc:HumanObservation	An observation made by one or more people.	none	dwctype:HumanObservation
dwc:MachineObservation	An observation made by a machine.	none	dwctype:MachineObservation
dwc:NomenclaturalChecklist	A nomenclatural checklist.	none	dwctype:NomenclaturalChecklist

